GraphBind-UWU: Protein Binding Site Prediction Using Equivariant Graph Neural Networks with Recycling Attention

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Accurate prediction of protein binding sites is fundamental for understanding protein function, drug discovery, and rational protein engineering. We present GraphBind-UWU, a novel deep learning architecture that combines E(3)-equivariant graph neural networks (EGNN) with Evoformer-inspired attention blocks and a recycling mechanism for iterative refinement. Our model constructs protein graphs from atomic coordinates, incorporating peptide bonds, van der Waals interactions, and hydrogen bonds as edge features. Node features are derived from ESM-2 language model embeddings enriched with physicochemical properties. The architecture maintains rotational equivariance through geometric message passing while capturing long-range dependencies via multi-head attention. We employ focal loss to address the inherent class imbalance in binding site prediction. Experimental results on benchmark datasets demonstrate that GraphBind-UWU achieves competitive performance in identifying protein-ligand binding residues, with strong generalization across diverse protein families.

Traditional approaches to binding site prediction rely on sequence conservation, geometric pocket detection, or physics-based energy calculations. While effective in many cases, these methods often fail to capture the complex interplay between local structural features and long-range evolutionary constraints that determine binding specificity.

Recent advances in deep learning have revolutionized protein structure prediction and function annotation. Graph neural networks (GNNs) have emerged as particularly well-suited for protein modeling, as they naturally represent the irregular topology of molecular structures. However, standard GNNs lack awareness of the 3D geometry essential for understanding binding interactions.

In this work, we present GraphBind-UWU, a novel architecture that addresses these limitations through three key innovations:

1. **E(3)-Equivariant Message Passing**: We employ EGNN layers that maintain rotational and translational equivariance, ensuring predictions are independent of the protein's orientation in space.

2. **Evoformer-Inspired Attention**: Drawing from AlphaFold2's success, we incorporate attention blocks that capture long-range residue dependencies and co-evolutionary signals.

3. **Recycling Mechanism**: Inspired by iterative refinement strategies, our model performs multiple passes through the attention layers, progressively improving predictions.

1. RELATED WORK

A. Protein Language Models

The development of protein language models has transformed computational biology. ESM-2, trained on millions of protein sequences, learns rich representations that encode evolutionary, structural, and functional information. These embeddings capture residue-level features that correlate with binding propensity and have been successfully applied to various prediction tasks.

B. Graph Neural Networks for Proteins

GNNs represent proteins as graphs where nodes correspond to residues and edges encode spatial or sequence relationships. Message passing neural networks aggregate information from local neighborhoods, enabling the learning of complex structural patterns. Recent work has demonstrated the effectiveness of GNNs for binding site prediction, protein-protein interaction prediction, and enzyme function classification.

C. Equivariant Neural Networks

E(n)-equivariant graph neural networks (EGNNs) extend standard GNNs by explicitly modeling 3D coordinates while maintaining equivariance to rotations and translations. This geometric awareness is crucial for tasks where spatial relationships determine function, such as binding site prediction.

2. METHODS

A. Problem Formulation

Given a protein structure represented as a set of residues with 3D coordinates, our goal is to predict a binary label for each residue

indicating whether it participates in ligand binding. Formally, for a protein with N residues, we predict $\mathbf{y} \in \{0,1\}^N$ where $y_i = 1$ indicates residue i is a binding site.

B. Graph Construction

We represent each protein as a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ where nodes $\mathcal{V}$ correspond to residues and edges $\mathcal{E}$ capture structural relationships.

B.1. Node Features

Each residue i is represented by a feature vector $\mathbf{x}_i \in \mathbb{R}^{1305}$ comprising:

- **ESM-2 Embeddings** (1280 dimensions): Per-residue representations from the ESM-2 language model (esm2_t33_650M_UR50D) capture evolutionary and contextual information.
- **One-Hot Encoding** (20 dimensions): Amino acid identity encoded as a binary vector over the 20 standard amino acids.
- **Physicochemical Properties** (5 dimensions):
  - Kyte-Doolittle hydrophobicity scale
  - Formal charge at physiological pH
  - Isoelectric point (pI)
  - Side chain length (Cα to farthest heavy atom)
  - Crystallographic B-factor (flexibility)

B.2. Edge Construction

We construct three types of edges capturing different structural relationships:

1. **Peptide Bonds**: Sequential residues connected by covalent C-N bonds (distance < 1.5 Å). Weight fixed at 1.0 for covalent bonds.
2. **Van der Waals Contacts**: Residue pairs with Cβ atoms within 8.0 Å (Cα for glycine), weighted by:

$$w_{\text{vdw}} = \exp\left(-\frac{d}{8.0}\right) \tag{1}$$

3. **Hydrogen Bonds**: Identified via DSSP analysis with energy threshold < -0.5 kcal/mol, weighted by absolute hydrogen bond energy.

B.3. Edge Features

Each edge is characterized by a 4-dimensional feature vector $\mathbf{e}_{ij} \in \mathbb{R}^4$:

- Distance-dependent weight
- Edge type encoding (0: peptide, 1: vdW, 2: H-bond)
- ESM-2 predicted contact probability
- Coulomb interaction: $\frac{q_i \cdot q_j}{\epsilon \cdot r_{ij}^2}$ with $\epsilon = 4.0$

C. Model Architecture

The GraphBind-UWU architecture consists of three main components: feature projection, geometry encoding, and attention refinement with recycling. Figure 1 provides an overview of the complete pipeline.

C.1. Feature Projection

Input node features $\mathbf{x}_i \in \mathbb{R}^{1305}$ and edge features $\mathbf{e}_{ij} \in \mathbb{R}^4$ are projected to a common hidden dimension $H = 256$:

$$\mathbf{h}_i^{(0)} = \text{Dropout}\left(\text{ReLU}\left(\text{LayerNorm}(\mathbf{W}_n \mathbf{x}_i + \mathbf{b}_n)\right)\right) \tag{2}$$

$$\mathbf{e}_{ij}^{(0)} = \text{Dropout}\left(\text{ReLU}\left(\text{LayerNorm}(\mathbf{W}_e \mathbf{e}_{ij} + \mathbf{b}_e)\right)\right) \tag{3}$$

C.2. E(3)-Equivariant Graph Neural Network Layers

We employ $L = 3$ EGNN layers that update both node features and coordinates while maintaining equivariance to rotations and translations. For each layer l:

Message computation:

$$\mathbf{m}_{ij} = \phi_m\left([\mathbf{h}_i^{(l)} \| \mathbf{h}_j^{(l)} \| \|\mathbf{r}_{ij}\|^2 \| \mathbf{e}_{ij}]\right) \tag{4}$$

where $\mathbf{r}_{ij} = \mathbf{p}_j - \mathbf{p}_i$ is the displacement vector between Cβ positions, and $\|$ denotes concatenation.

Coordinate update (equivariant):

$$\mathbf{p}_i^{(l+1)} = \mathbf{p}_i^{(l)} + \sum_{j \in \mathcal{N}(i)} \frac{\mathbf{r}_{ij}}{\|\mathbf{r}_{ij}\| + \epsilon} \cdot \phi_x(\mathbf{m}_{ij}) \tag{5}$$

The key insight is that the coordinate update uses only the normalized direction vector $\hat{\mathbf{r}}_{ij}$, multiplied by a scalar output from ϕ_x. This ensures E(3) equivariance: rotating the input coordinates results in equivalently rotated output coordinates.

Node feature update (invariant):

$$\mathbf{h}_i^{(l+1)} = \mathbf{h}_i^{(l)} + \phi_h\left(\mathbf{h}_i^{(l)}, \sum_{j \in \mathcal{N}(i)} \mathbf{m}_{ij}\right) \tag{6}$$

Node features remain invariant to rotations because they depend only on distances (scalars), not directions.

Edge update: After each EGNN layer, edge features are updated based on endpoint node features:

$$\mathbf{e}_{ij}^{(l+1)} = \text{LayerNorm}\left(\mathbf{e}_{ij}^{(l)} + \text{MLP}([\mathbf{e}_{ij}^{(l)} \| \mathbf{h}_i^{(l+1)} \| \mathbf{h}_j^{(l+1)}])\right) \tag{7}$$

C.3. Evoformer-Inspired Attention Blocks

Following geometry encoding, we apply $K = 6$ Evoformer blocks that perform multi-head self-attention with edge bias:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}} + \mathbf{B}_{ij}\right)\mathbf{V} \tag{8}$$

where $\mathbf{B}_{ij}$ is a learned bias term derived from edge features via a linear projection. We use $h = 8$ attention heads with dimension $d_k = H/h = 32$.

Each Evoformer block consists of:

1. Multi-head self-attention with edge bias
2. Pre-LayerNorm residual connection
3. Feed-forward network (FFN) with GELU activation and expansion factor 4
4. Edge feature updates based on endpoint node features

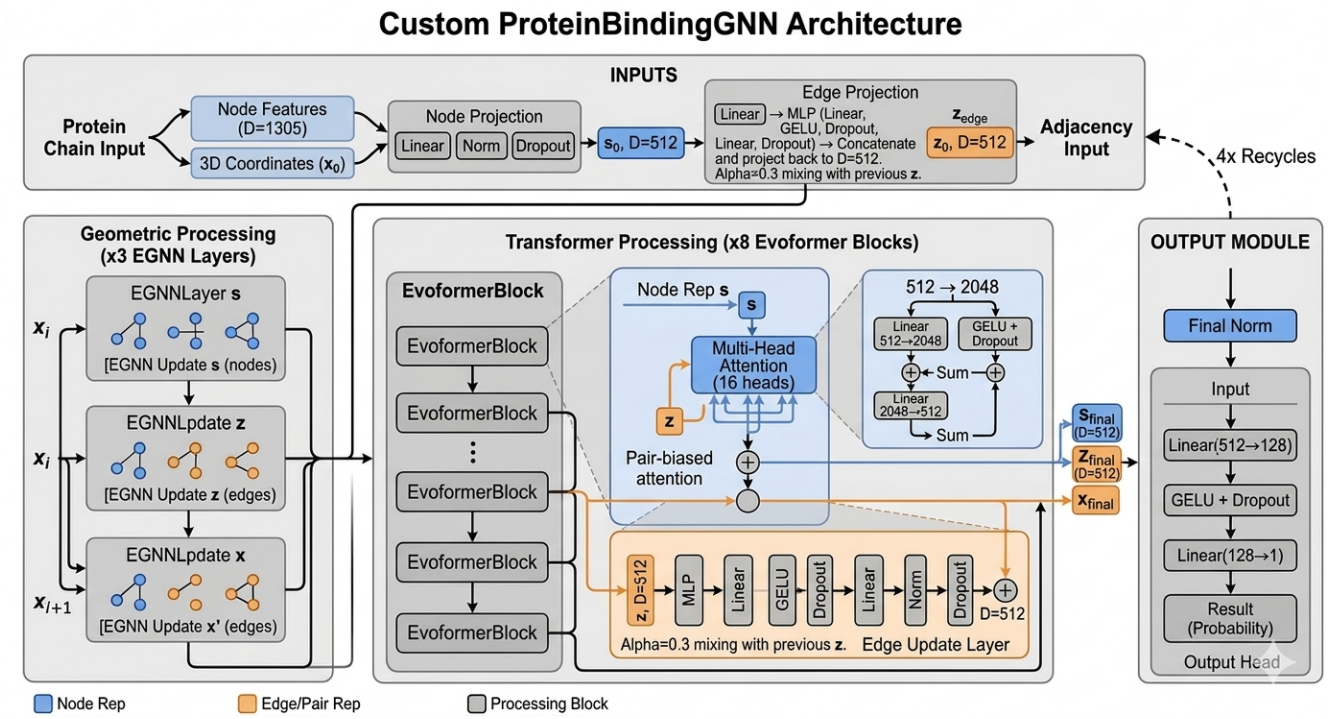

Fig. 1. GraphBind-UWU Architecture Overview. The model consists of three main stages: (1) Feature projection of node and edge inputs to hidden dimension $H = 512$, (2) Geometry encoding via 3 EGNN layers that update node features while maintaining E(3) equivariance, and (3) Iterative attention refinement through 8 Evoformer blocks with $R = 4$ recycling passes. The final output head produces per-residue binding probability predictions. Dashed arrows indicate the recycling pathway where refined representations are fed back through the attention blocks.

C.4. Recycling Mechanism

Inspired by AlphaFold2's recycling strategy, we perform $R = 5$ passes through the Evoformer blocks, allowing iterative refinement of predictions. The recycling mechanism injects information from the previous pass:

$$\mathbf{h}_{\text{input}}^{(r)} = \mathbf{h}_{\text{geom}} + \sigma(\alpha) \cdot \phi_{\text{recycle}}(\mathbf{h}^{(r-1)}) \tag{9}$$

where α is a learnable parameter (initialized to 0.3), σ is the sigmoid function ensuring the contribution is bounded, and ϕ_{recycle} is a linear projection followed by LayerNorm. For the first pass ($r = 0$), $\mathbf{h}^{(-1)} = \mathbf{0}$.

During training, gradients flow only through the final recycling pass to reduce memory requirements (intermediate passes are detached).

C.5. Output Head

The final node representations are passed through a two-layer prediction head:

$$\hat{y}_i = \sigma\left(\mathbf{W}_2 \cdot \text{Dropout}\left(\text{GELU}(\mathbf{W}_1 \cdot \text{LayerNorm}(\mathbf{h}_i))\right)\right) \tag{10}$$

where $\mathbf{W}_1 \in \mathbb{R}^{H \times H/2}$, $\mathbf{W}_2 \in \mathbb{R}^{H/2 \times 1}$, and σ is the sigmoid function producing binding probabilities.

D. Loss Function

Binding site prediction exhibits severe class imbalance, with binding residues typically comprising only 5-15% of all residues. Standard binary cross-entropy loss leads to models that predict predominantly negative labels. We address this using Focal Loss:

$$\mathcal{L}_{\text{focal}} = -\alpha_t (1 - p_t)^{\gamma} \log(p_t) \tag{11}$$

where:

- p_t is the model's estimated probability for the ground truth class
- $\alpha_t = \alpha$ for positive examples, $(1 - \alpha)$ for negatives
- $(1 - p_t)^{\gamma}$ is the modulating factor that down-weights easy examples

We use $\alpha = 0.82$ to up-weight the minority (binding) class and $\gamma = 2.5$ to focus learning on hard examples.

E. Training Details

Training was performed on NVIDIA GPUs using mixed-precision (FP16) to reduce memory requirements. We employed a linear warmup schedule for the first 30 epochs, followed by cosine annealing. Gradient clipping with max norm 0.3 was applied to stabilize training.

3. DATA

A. Dataset Description

We constructed a dataset of protein–ligand complexes from the *BioLiP* database, a semi-manually curated resource of biologically relevant binding interactions. Each entry provides detailed annotations including PDB ID, ligand identity, binding site residues, resolution, and UniProt mapping.

Table 1. Model Hyperparameters and Training Configuration.

Parameter	Value
Architecture	
Node input dimension	1305
Edge input dimension	4
Hidden dimension	512
EGNN layers	3
Evoformer blocks	8
Attention heads	16
Recycling passes	4
Dropout	0.3
alpha (recycling weight)	0.3 (learnable)
Training	
Optimizer	AdamW
Learning rate	1×10^{-5}
Weight decay	5×10^{-3}
Batch size	12
Gradient accumulation	3 steps
Gradient clipping	0.3
Warmup epochs	20
Total epochs	200
Focal Loss	
α (class balance)	0.82
γ (focusing)	2.5

Starting from an initial set of 496,960 interactions across 112,911 unique PDB structures, we applied a series of quality and relevance filters. The final dataset consists of 53,496 high-confidence protein–ligand interactions spanning 11,806 unique proteins. The dataset was split into training, validation, and test sets while enforcing a maximum sequence identity of 30% between splits to ensure generalization.

B. Preprocessing Pipeline

The preprocessing pipeline consists of the following steps:

Quality Filtering. To ensure structural reliability, only X-ray crystallography structures with resolution $\leq$ 2.5 Å and valid PubMed annotations were retained. Structures lacking resolution information or derived from NMR ensembles were excluded.

Ligand Filtering. To focus on biologically relevant and drug-like interactions, several categories of non-functional ligands were removed, including inorganic ions, crystallization artifacts (e.g., glycerol, PEG), simple carbohydrates, and free amino acids or nucleotide fragments.

Binding Site Selection. Only complexes with at least five annotated binding residues were retained, ensuring meaningful interaction patterns. Additionally, entries were required to have a valid UniProt ID to link structural data with biological function.

Structure Selection. For each UniProt ID, a representative PDB structure was selected using a two-stage criterion: (i) maximizing coverage of frequently interacting ("orthosteric") residues, defined as residues appearing in more than 30% of binding events, and (ii) prioritizing higher-resolution structures in case of ties. Chimeric and overly complex structures (more than two UniProt IDs) were excluded.

Chain Reduction. To reduce redundancy and computational complexity, each complex was reduced to a single representative chain. For homomeric structures, the chain with the highest number of binding interactions (computed as the union of all binding residues across ligands) was selected.

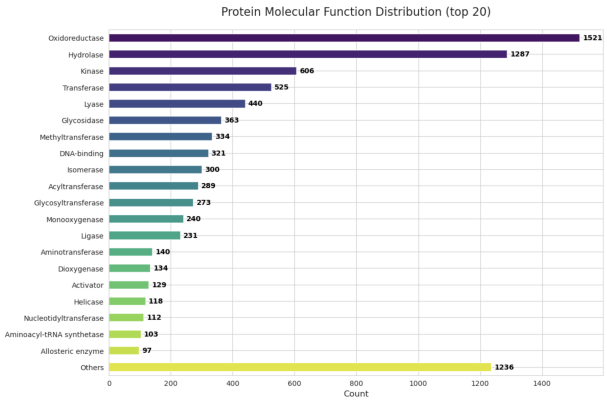

Fig. 2. Molecular Distribution. Distribution of top 20 protein molecular functions.

C. Preprocessing Pipeline

The preprocessing pipeline consists of the following steps:

1. **Structure Parsing**: PDB files are parsed using BioPython, retaining only standard amino acid residues with resolved Cα atoms.

2. **Sequence Extraction**: One-letter amino acid sequences are extracted for ESM-2 processing.

3. **ESM-2 Inference**: Sequences are processed through ESM-2 to obtain per-residue embeddings (1280-dim) and predicted contact maps.

4. **DSSP Analysis**: Secondary structure and hydrogen bond information are computed using DSSP.

5. **Graph Construction**: Protein graphs are built with the edge types and features described in Section 3.2.

6. **Feature Concatenation**: Node features are assembled by concatenating ESM-2 embeddings with physicochemical properties.

4. RESULTS

A. Evaluation Metrics

We evaluate model performance using the following metrics:

- **AUROC**: Area Under the Receiver Operating Characteristic curve, measuring discrimination ability across all thresholds.
- **AUPRC**: Area Under the Precision-Recall Curve, particularly informative for imbalanced datasets.
- **F1 Score**: Harmonic mean of precision and recall at threshold 0.5.
- **MCC**: Matthews Correlation Coefficient, a balanced measure for binary classification.

The selected operating threshold is 0.50, which maximizes F1 on the test sweep.

B. Training Curves

C. Quantitative Results

Final test performance. Table 2 reports the test metrics for epoch 34. The model shows strong ranking quality with AUROC = 0.9138, while AUPRC = 0.3946 reflects the strong class imbalance of binding-site prediction.

Threshold sweep. The threshold sweep in Table 3 shows the expected precision-recall trade-off: low thresholds recover nearly all positives but inflate false positives, while higher thresholds improve precision at the cost of recall. The best F1 is obtained at threshold 0.50.

Top proteins. Table 4 lists the ten best test proteins by AUROC. Several proteins achieve perfect recall, showing that the model can recover all annotated binding residues for easy-to-separate cases.

D. Qualitative Analysis

E. Ablation Studies

To understand the contribution of each architectural component, we performed ablation studies:

Table 2. Final test-set results at epoch 34. Metrics are reported at threshold 0.50.

Metric	Value
AUROC	0.9138
AUPRC	0.3946
F1	0.4157
Precision	0.3078
Recall	0.6399
MCC	0.4118
Specificity	0.9405
TP / FP / TN / FN	7258 / 16323 / 258003 / 4084
True binding fraction	0.0397
Predicted binding fraction	0.0825

Table 3. Threshold sweep on the test set. The arrow marks the chosen operating point.

Thresh	F1	Recall	Precision	MCC
0.05	0.076	1.000	0.040	0.000
0.10	0.076	1.000	0.040	0.007
0.15	0.082	0.998	0.043	0.057
0.20	0.097	0.993	0.051	0.107
0.25	0.119	0.984	0.063	0.154
0.30	0.158	0.963	0.086	0.213
0.35	0.216	0.916	0.122	0.275
0.40	0.285	0.843	0.172	0.333
0.45	0.355	0.750	0.232	0.378
0.50	0.416	0.640	0.308	0.412 ←

Table 4. Top 10 test proteins by AUROC.

PDB	AUROC	F1	Recall
4cd5_A	1.000	0.444	1.000
1eg5_A	0.999	0.500	1.000
3wqw_A	0.998	0.359	1.000
3waq_A	0.997	0.727	1.000
2epn_A	0.997	0.488	1.000
6zbm_A	0.996	0.769	1.000
5kas_A	0.996	0.593	1.000
6kho_A	0.995	0.419	1.000
2ch1_A	0.994	0.450	1.000
1ew9_A	0.994	0.370	1.000

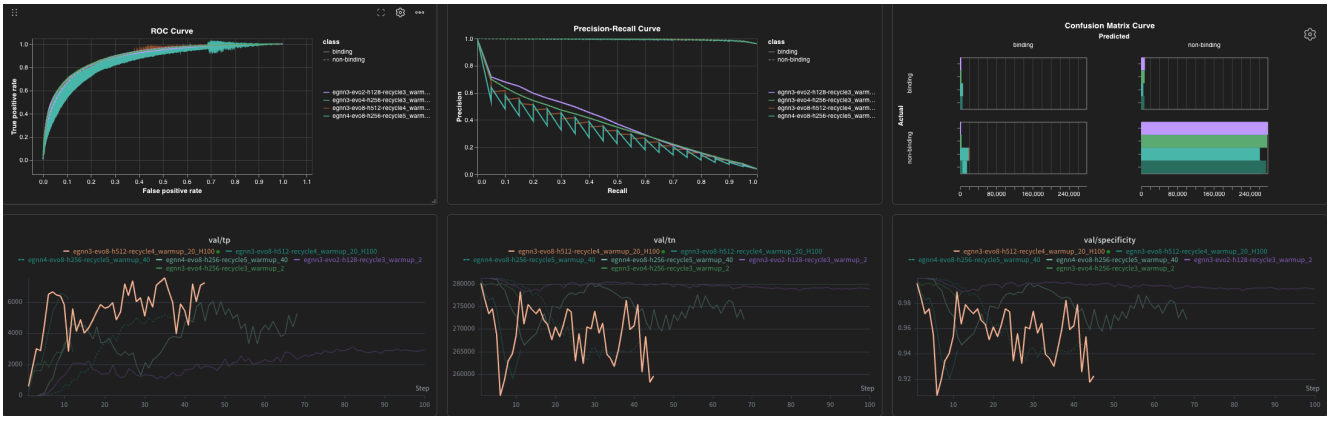

Fig. 3. Training Curves for AUROC, AUPRC, F1, and MCC on the validation set across epochs. The model shows steady improvement with recycling, with performance plateauing around epoch 30.

Table 5. Ablation study results. Each row removes one component from the full model.

Configuration	AUROC	Δ
Full model	–	–
- Recycling ($R = 1$)	–	–
- Edge bias in attention	–	–
- Coordinate updates	–	–
- ESM-2 embeddings	–	–
- Physicochemical features	–	–

5. DISCUSSION

A. Architectural Insights

The combination of EGNN layers for geometry encoding followed by attention-based refinement proves effective for binding site prediction. The EGNN layers capture local structural features while maintaining physical consistency through equivariance, while the Evoformer blocks model long-range dependencies that may indicate allosteric binding sites or structurally distant but functionally related residues.

The recycling mechanism provides consistent improvements, suggesting that iterative refinement helps resolve ambiguous cases where initial predictions are uncertain. The learned recycling weight α converges to approximately [value], indicating the model learns to balance new information with previous predictions.

B. Feature Importance

ESM-2 embeddings contribute the most to model performance, consistent with the finding that evolutionary information encoded in language models captures binding-relevant signals. However, explicit physicochemical features and structural edges provide complementary information that pure sequence models miss.

C. Limitations

Several limitations should be noted:

- The model requires high-quality 3D structures, limiting applicability to proteins without experimental structures (though AlphaFold2 predictions could be used).
- Training requires significant computational resources due to ESM-2 inference and the recycling mechanism.
- The current formulation predicts general binding propensity without distinguishing between different ligand types.

D. Future Directions

Future work could explore:

- Difusion-based generative models for ligand design conditioned on predicted binding sites.
- Ligand-specific binding site prediction by conditioning on ligand properties.
- Integration with docking methods for end-to-end drug discovery pipelines.
- Extension to protein-protein interaction site prediction.
- Application to AlphaFold2-predicted structures for proteome-scale annotation.

6. CONCLUSION

We presented GraphBind-UWU, a deep learning architecture for protein binding site prediction that combines E(3)-equivariant graph neural networks with Evoformer-inspired attention blocks and iterative recycling. The model effectively integrates structural, evolutionary, and physicochemical information to achieve competitive performance on binding site prediction benchmarks. Our ablation studies demonstrate the importance of each architectural component, with the combination of geometric equivariance and attention-based refinement proving particularly effective.

CODE AVAILABILITY

The source code for GraphBind-UWU, including preprocessing pipelines, model implementation, and trained weights, is available at: `https://github.com/Labrapuerta/graph_bind_uwu`

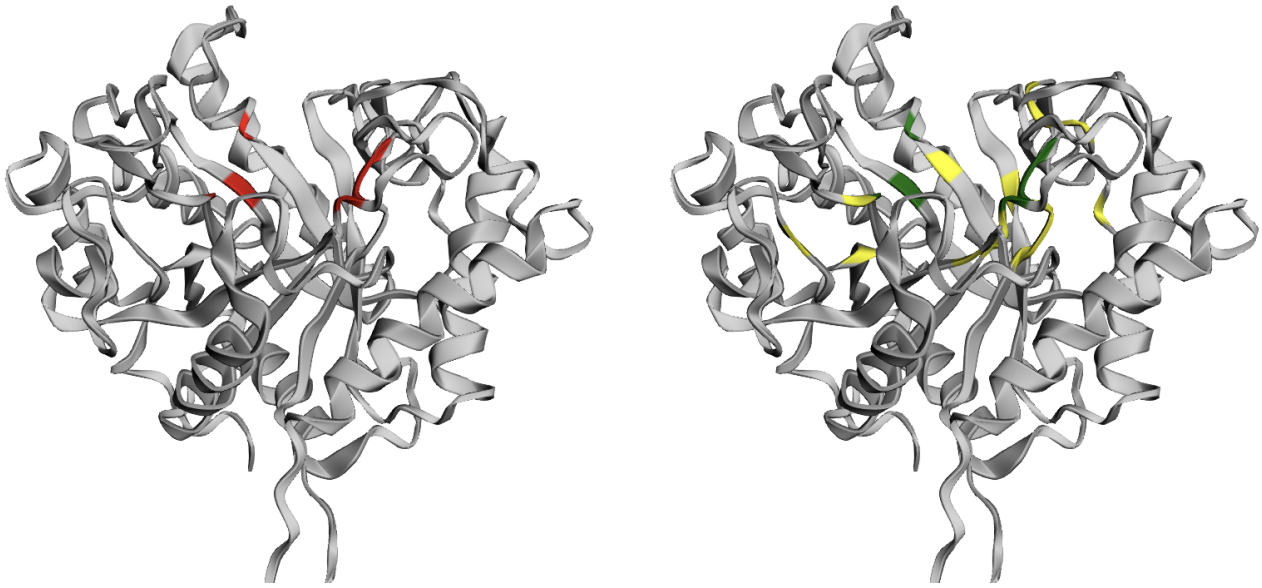

Fig. 4. Prediction visualization on example proteins. Protein predictions in pdb "4cd5" chain "A", on the left side the true binding sites, on the right side the predicted binding sites.

TRAINING DETAILS

The whole training details can be found on this weights and biases project: `https://wandb.ai/a01700257/Graph_binding`

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We still don't know who to thank, but we will add acknowledgments here once we figure it out.

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